

(a) Coordinate tree, $h = 3$



Occupied: 001, 100, 110. Target: 001.

(b) Proof for 001

Level 0: Δ_0 local

Level 1: Δ_1 local

Level 2: R retrieved

Proof: $[\Delta_0, \Delta_1, R]$

(c) Fixed batch: one query per color ($m = 2$)

Color 1: real PIR query

R $\rightarrow$ original level 2

Color 2: dummy PIR

Discard its response

Real/dummy roles remain client-local.

Known value + coordinate bits 001

+ proof $[\Delta_0, \Delta_1, R] \rightarrow$ computed root

Accept iff computed root = trusted root